

## 2 Biology - The Life Science

The word **biology** means, "the science of life", from the Greek **bios**, *life*, and **logos**, *word* or *knowledge*. Therefore, Biology is the science of Living Things. That is why Biology is sometimes known as Life Science.

The science has been divided into many subdisciplines, such as botany<sup>1</sup>, bacteriology, anatomy<sup>2</sup>, zoology, histology, mycology, embryology, parasitology, genetics<sup>3</sup>, molecular biology<sup>4</sup>, systematics, immunology, microbiology<sup>5</sup>, physiology, cell biology<sup>6</sup>, cytology, ecology<sup>7</sup>, and virology. Other branches of science include or are comprised in part of biology studies, including paleontology<sup>8</sup>, taxonomy, evolution, phycology, helimentology, protozoology, entomology, biochemistry, biophysics, biomathematics, bio engineering, bio climatology and anthropology.

### 2.1 Characteristics of life

Not all scientists agree on the definition of just what makes up life. Various characteristics describe most living things. However, with most of the characteristics listed below we can think of one or more examples that would seem to break the rule, with something nonliving being classified as living or something living classified as nonliving. Therefore we are careful not to be too dogmatic in our attempt to explain which things are living or nonliving.

- Living things are composed of **matter structured in an orderly way** where simple molecules are ordered together into much larger macromolecules.

An easy way to remember this is GRIMNERD C All organisms; - **G**row, **R**espire, **I**nteract, **M**ove, **N**eed **N**utrients, **E**xcrete (Waste), **R**eproduce, **D**eath, **C**ells (Made of)

- Living things are **sensitive**, meaning they are able to **respond to stimuli**.
- Living things are able to **grow, develop, and reproduce**.
- Living things are able to **adapt** over time by the process of **natural selection**.
- All known living things use the **hereditary molecule, DNA**<sup>9</sup>.

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1 <http://en.wikibooks.org/wiki/botany>  
2 <http://en.wikibooks.org/wiki/anatomy>  
3 <http://en.wikibooks.org/wiki/genetics>  
4 <http://en.wikibooks.org/wiki/Molecular%20Biology>  
5 <http://en.wikibooks.org/wiki/microbiology>  
6 <http://en.wikibooks.org/wiki/Cell%20Biology>  
7 <http://en.wikibooks.org/wiki/ecology>  
8 <http://en.wikibooks.org/wiki/paleontology>  
9 <http://en.wikipedia.org/wiki/DNA>

- Internal functions are coordinated and **regulated** so that the internal environment of a living thing is relatively constant, referred to as **homeostasis**<sup>10</sup>.

Living things are organized in the microscopic level from atoms up to cells<sup>11</sup>. Atoms are arranged into molecules, then into macromolecules<sup>12</sup>, which make up organelles<sup>13</sup>, which work together to form cells. Beyond this, cells are organized in higher levels to form entire multicellular organisms. Cells together form tissues<sup>14</sup>, which make up organs, which are part of organ systems, which work together to form an entire organism. Of course, beyond this, organisms form populations which make up parts of an ecosystem. All of the Earth's ecosystems together form the diverse environment that is the earth.

Example:-

sub atoms, atoms, molecules, cells, tissues, organs, organ systems, organisms, population, community, eco systems

## 2.2 Nature of science

Science is a **methodology** for **learning about the world**. It involves the **application of knowledge**.

The scientific method deals with **systematic investigation**, **reproducible results**, the formation and testing of **hypotheses**, and **reasoning**.

Reasoning can be broken down into two categories, **induction** (specific data is used to develop a generalized observation or conclusion) and **deduction** (general information leads to specific conclusion). Most reasoning in science is done through induction.

Science as we now know it arose as a discipline in the 17th century.

## 2.3 Scientific method

The scientific method is not a step by step, linear process. It is an intuitive process, a methodology for learning about the world through the application of knowledge. Scientists must be able to have an "imaginative preconception" of what the truth is. Scientists will often observe and then hypothesize the reason why a phenomenon occurred. They use all of their knowledge and a bit of imagination, all in an attempt to uncover something that might be true. A typical scientific investigation might go like so:

You *observe* that a room appears dark, and you ponder *why* the room is dark. In an attempt to find explanations to this curiosity, your mind unravels several different *hypotheses*. One hypothesis might state that the lights are turned off. Another hunch might be that the room's lightbulb has burnt out. Worst yet, you could be going blind. To discover the truth,

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10 <http://en.wikipedia.org/wiki/homeostasis>

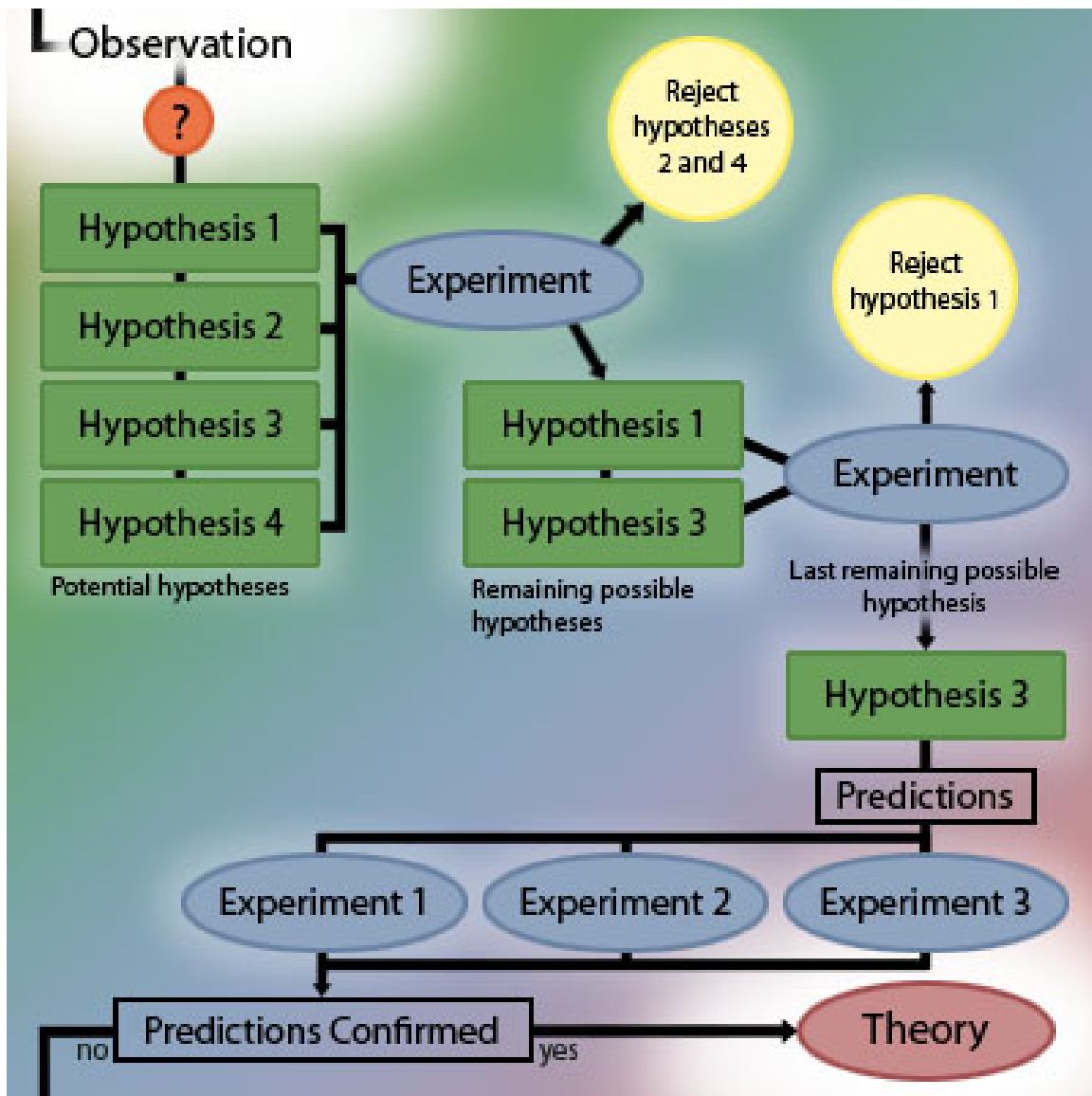
11 <http://en.wikipedia.org/wiki/cell>

12 <http://en.wikipedia.org/wiki/macromolecule>

13 <http://en.wikipedia.org/wiki/organelle>

14 <http://en.wikibooks.org/wiki/General%20Biology%2FTissues>

you *experiment*. You feel your way around the room and find a light switch and turn it on. No light. You *repeat* the experiment, flicking the switch back and forth. Still nothing. That means your initial hypothesis, the room is dark because the lights are off, has been *rejected*. You devise more experiments to test your hypotheses, utilizing a flashlight to prove that you are indeed not blind. In order to *accept* your last remaining hypothesis as the truth, you could *predict* that changing the light bulb will fix the problem. If all your predictions succeed, the original hypothesis is valid and is accepted. In some cases, however, your predictions will not occur, in which you'll have to start over. Perhaps the power is off.



**Figure 1 How Science is Done**

A diagram that illustrates scientific investigation

Scientists first make observations that raise a particular question. In order to explain the observed phenomenon, they develop a number of possible explanations, or hypotheses. This is the inductive part of science, observing and constructing plausible arguments for why

an event occurred. Experiments are then used to eliminate one of more of the possible hypotheses until one hypothesis remains. Using deduction, scientists use the principles of their hypothesis to make predictions, and then test to make sure that their predictions are confirmed. After many trials (repeatability) and all predictions have been confirmed, the hypothesis then may become a theory.

### Quick Definitions

**Observation** - Quantitative and qualitative measurements of the world.

**Inference** - Deriving new knowledge based upon old knowledge.

**Hypotheses** - A suggested explanation.

**Rejected Hypothesis** - An explanation that has been ruled out through experimentation.

**Accepted Hypothesis** - An explanation that has not been ruled out through excessive experimentation and makes verifiable predictions that are true.

**Experiment** - A test that is used to rule out a hypothesis or validate something already known.

**Scientific Method** - The process of scientific investigation.

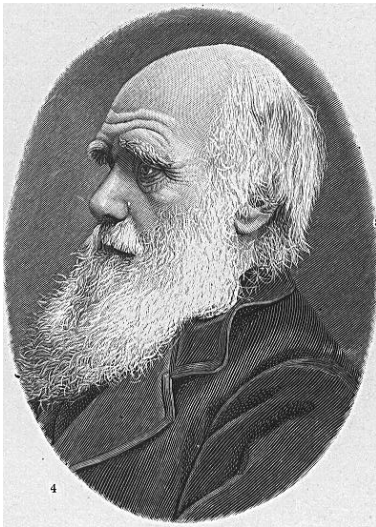
**Theory** - A widely accepted hypothesis that stands the test of time. Often tested, and usually never rejected.

The scientific method is based primarily on the testing of hypotheses by experimentation. This involves a **control**, or subject that does not undergo the process in question. A scientist will also seek to limit variables to one or another very small number, single or minimum number of variables. The procedure is to form a hypothesis or prediction about what you believe or expect to see and then do everything you can to violate that, or falsify the hypotheses. Although this may seem unintuitive, the process serves to establish more firmly what is and what is not true.

A founding principle in science is a lack of absolute truth: the accepted explanation is the most likely and is the basis for further hypotheses as well as for falsification. All knowledge has its relative uncertainty.

**Theories** are hypotheses which have withstood repeated attempts at falsification. Common theories include evolution by natural selection and the idea that all organisms consist of cells. The scientific community asserts that much more evidence supports these two ideas than contradicts them.

## 2.4 Charles Darwin



**Figure 2**

Charles Darwin is most remembered today for his contribution of the theory of **evolution through natural selection**.

The seeds of this theory were planted in Darwin's mind through observations made on a five-year voyage through the New World on a ship called the Beagle. There, he studied fossils and the geological record, geographic distribution of organisms, the uniqueness and relatedness of island life forms, and the affinity of island forms to mainland forms.

Upon his return to England, Darwin pondered over his observations and concluded that evolution must occur through natural selection. He declined, however, to publish his work because of its controversial nature. However, when another scientist, Wallace, reached similar conclusions, Darwin was convinced to publish his observations in 1859. His hypothesis revolutionized biology and has yet to be falsified by empirical data collected by mainstream scientists.

## 2.5 After Darwin

Since Darwin's day, scientists have amassed a more complete **fossil record**, including **microorganisms** and **chemical fossils**. These fossils have supported and added subtleties to Darwin's theories. However, the age of the Earth is now held to be much older than Darwin thought. Researchers have also uncovered some of the preliminary mysteries of the mechanism of heredity as carried out through **genetics** and **DNA**, areas unknown to Darwin. Another growing area is **comparative anatomy** including homology and analogy.

Today we can see a bit of evolutionary history in the **development** of embryos, as certain (although not all) aspects of development recapitulate evolutionary history.

The molecular biology<sup>15</sup> study of slowly mutating genes reveal considerable evolutionary history consistent with fossil and anatomical record.

## 2.6 Challenges to Darwin



**Figure 3**

Darwin and his theories have been challenged many times in the last 150 years. The challenges have been primarily religious based on a perceived conflict with the preconceived notion of creationism. Many of those who challenge Darwin have been adherents to the young earth hypothesis that says that the Earth is only some 6000 years old and that all species were individually created by a god. Some of the proponents of these theories have suggested that chemical and physical laws that exist today were different or nonexistent in earlier ages. However, for the most part, these theories are either not scientifically testable and fall outside the area of attention of the field of biology, or have been disproved by one or more fields of science.

*This text is based on notes very generously donated by Dr. Paul Doerder, Ph.D., of Cleveland State University.*

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<sup>15</sup> <http://en.wikibooks.org/wiki/molecular%20biology>